LESSON PLAN

AIR WEAPON (10)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Drag analysis, induced, form, skin friction, profile parasite and total drag
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Drag analysis, induced, form, skin friction, profile parasite and total drag.
5. Specific Objectives	10.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer and various concepts of Aerodynamics.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review drag component diagrams; verify definitions of induced, form, skin friction, profile parasite, and total drag; instructor supervises concept mapping.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the four primary forces acting on an aircraft in flight?
(b) Define the boundary layer in the context of airflow over a body.
(c) List the three main components of parasite drag.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Drag and its significance • Skin friction drag and boundary layer effects • Form drag and pressure distribution • Profile drag as sum of skin friction and form • Induced drag and lift-induced downwash	L & Ds	PPT, IB	15 min
(b)	• Parasite drag components breakdown • Relationship between speed and drag types • Total drag curve analysis • Minimum drag speed determination • Impact of airfoil shape on drag	L & Ds	PPT, PDF	20 min
(c)	• Comparison of induced and parasite drag • Operational implications of drag analysis • Fuel efficiency and drag management • Reinforcement of total drag concept • Summary of drag types and causes	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how induced drag changes as airspeed increases.
(b) Differentiate between form drag and skin friction drag with examples.
(c) Describe the relationship between total drag and the minimum drag speed.

12. Home Assignment and Follow Up	Prepare a brief note on how understanding drag components aids in optimizing aircraft performance for the next lesson on lift optimization.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Sub-Lieutenant, P No. 12345	Senior Instructor, Lieutenant, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min